

Blue Ember Damages Expert Report

Damages analysis for Blue Ember Biologics, LLC

Civil Action No. SYN-WDNC-24-CV-0520

Report dated July 7, 2025

I was retained by Holt Bell PLLC to evaluate economic loss claimed by Blue Ember Biologics, LLC arising from the September 12, 2024 temperature excursion affecting validation lot BE214-VL-2409A. My opinions assume the factfinder determines contractual responsibility and causation; I do not decide breach, product release, or the governing mitigation instruction.

My principal opinion is that Blue Ember's gross loss is 4.2 million dollars. That amount comprises 2.2 million dollars of direct lot and replacement cost and 2 million dollars of bounded delay loss. The components are distinct, draw from different sources, and can be separately adjusted without changing liability.

I also reviewed the October 2 Cobalt Finch capacity offer. It represented a genuine possible alternative, but the evidence available by the October 4 deadline did not make a successful December campaign certain enough to impose Granite Heron's proposed 1.5 million dollar deduction as an economic fact. I provide sensitivities so the factfinder can apply its own findings without duplicating a cost or avoidance amount.

Qualifications and compensation

I am a certified public accountant and hold the Certified in Financial Forensics credential. During twenty-one years of fictional professional practice, I have analyzed project cost, supply interruption, manufacturing delay, contribution margin, mitigation, and damages documentation in life-science and specialty-manufacturing disputes.

My work includes tracing general-ledger activity to source invoices, evaluating pre-event forecasts, separating incremental from ordinary cost, and testing alternative schedules. I have taught continuing-education courses on uncertainty in development-stage loss and on avoiding duplication between replacement cost and time-based models.

Holt Bell PLLC compensates me at 525 dollars per hour for analysis and 650 dollars per hour for deposition or trial testimony. My compensation does not depend on outcome or amount. I have billed approximately 38,000 dollars through the report date and expect additional time for testimony and rebuttal.

A fictional curriculum vitae appended to this report lists publications and testimony from the preceding four years. I have testified four times at deposition and twice at trial during that period. No court has excluded my opinions in this training record. Those background statements establish qualifications only; they do not substitute for examination of my methods here.

Assignment and opinion boundaries

Counsel asked me to identify economic loss supported by Blue Ember's records, distinguish direct expense from delay, test whether forecasts predated the event, evaluate credits and ordinary costs, and assess the financial implications of the Cobalt Finch proposal. I was also asked to show calculations in a form that permits a different factual finding to be inserted.

I assume the shipment remained outside its required range from 13:08 through 14:02 and that Blue Ember rejected the validation lot on September 16. I assume the quality decision was made in good faith. Whether rejection was contractually caused is for other evidence and the factfinder.

I do not offer a regulatory, biochemical, carrier-practice, or legal opinion. I do not say that the assay proved every vial inactive. I do not interpret the words "foreseeable," "reasonable mitigation," or "recoverable" as legal conclusions. My economic analysis is conditional on the Court's instructions.

The report values a defined loss period rather than perpetual future sales. It does not calculate punitive damages, reputational harm, enterprise-value diminution, attorney fees, prejudgment interest, or a loss after the stated program horizon.

Information reviewed

I reviewed the master agreement, quality agreement, statement of work, purchase order, release specification, rate confirmation, custody and CTX-9 materials, deviation and disposition records, assay summary, replacement campaign records, invoices, general-ledger extracts, budgets, forecasts, and program calendars.

I considered deposition testimony from Dr. Mira Solis, Adrian Locke, Tess Holloway, and Dr. Nia Calder. I reviewed the October 2 Cobalt Finch offer, question log, preliminary transfer plan, cost estimate, capacity calendar, and Blue Ember's nonprivileged decision assessment. I compared that evidence with the March 3, 2025 Aurora Quay schedule.

Financial data were reconciled to project codes and sampled invoices. I compared the claimed budget with versions created before September 12 to reduce hindsight selection. Where a source used a range, I retained the range until a stated assumption was required.

Counsel supplied pleadings and discovery responses to identify disputed theories, but I did not assume an allegation was true merely because it appeared in a pleading. My workpapers identify source, date, adjustment, and classification. If material records are supplemented, I will evaluate whether an opinion changes and disclose a timely revision.

Economic timeline

The relevant event sequence begins with Granite Heron's September 12 custody and the temperature curve. Blue Ember quarantined the lot on delivery and rejected it September 16. The rejection created an immediate asset and replacement issue, followed by a schedule-choice issue.

Cobalt Finch offered held capacity on October 2, subject to acceptance by October 4 at 17:00. Its preliminary plan called for an October 7 kickoff, October 14–15 site work, November engineering confirmation, a November 22 readiness decision, and production beginning December 2. The successful path contemplated initial campaign completion near December 20, with later testing and disposition.

Blue Ember declined that course and continued with the more established Aurora Quay facility. The replacement campaign began March 3, 2025. My delay model measures the bounded difference created by displacement of the planned development sequence, subject to program-specific risk adjustments.

Direct cost is not calculated by multiplying months. It arises from the rejected lot and duplicated replacement effort. Delay loss is not calculated by adding the same invoices again. Keeping the chronology and sources separate permits the factfinder to accept one component, reject another, or apply a mitigation deduction without disturbing the liability question.

Cost embodied in validation lot

Blue Ember's project ledger attributes 1.35 million dollars to BE214-VL-2409A before the transport event. The amount includes 760,000 dollars in contracted production and processing, 245,000 dollars in materials and consumables, 185,000 dollars in analytical and release services, and 160,000 dollars in campaign-specific internal labor and support.

I traced the external categories to invoices, purchase records, and vendor accruals. I tested internal allocations against the method Blue Ember used for other BE-214 campaigns before the loss. I excluded a broad corporate-overhead loading that was not tied to the lot.

The total equals the enhanced declared payload value, but I did not infer cost solely from the declaration. The underlying ledger independently reconciles to 1.35 million dollars. Declared value is corroborative of contemporaneous scale and risk awareness.

I found no material salvage credit. The lot remained segregated, provided limited investigation samples, and was not used for validation or a revenue-producing purpose. If evidence establishes a quantifiable alternative use, that value should reduce this component once. Controlled destruction cost appears in the replacement category and is not embedded here.

Replacement and directly associated expense

The additional direct component is 850,000 dollars. It consists of 555,000 dollars in replacement manufacturing services, 135,000 dollars in repeated analytical and release work, 90,000 dollars in incremental materials and logistics, 45,000 dollars in campaign-specific technical labor, and 25,000 dollars in disposition and controlled-destruction services.

I tested invoices against project codes and removed ordinary forward work that would have occurred even if the September lot had conformed. I also removed 70,000 dollars initially included for broad management time because contemporaneous records did not establish incremental effort with adequate specificity.

The 850,000 dollars does not include the original 1.35 million dollar lot cost. Nor does it include the 300,000-dollar Cobalt Finch estimate because Blue Ember did not accept or incur that alternative. A factfinder applying a hypothetical alternative must substitute the relevant cost in the mitigation calculation rather than add it to actual expense.

No insurance receipt or vendor credit duplicating these items appears through the report date. Blue Ember has a continuing obligation to update that information. If a later payment compensates the same invoice, the corresponding amount should be credited, subject to whatever legal treatment the Court directs.

Delay-loss method

I calculate 2 million dollars of delay loss using a defined schedule window and contribution approach derived from pre-event program plans. The model begins with milestone timing Blue Ember used in its August 2024 operating forecast, then shifts the validation path to reflect the replacement campaign and applies risk-weighted contribution associated with the affected interval.

The model does not assume immediate market launch, unlimited sales, or certainty of regulatory approval. It uses contractual collaboration and development economics that were already incorporated into Blue Ember's planning. Later commercial possibilities outside the bounded period are excluded.

I compared forecast versions, removed a post-event upside scenario, and used the lower of two contemporaneous volume assumptions. I applied the program risk factor used internally before the loss rather than a litigation-created success rate. Avoided variable spending was deducted from the gross milestone and contribution effect.

The resulting figure is 2.04 million dollars, rounded down to 2 million to avoid false precision. The workpaper permits the timing shift, risk factor, or contribution assumption to be changed independently. The amount remains distinct from the costs already counted in the 2.2 million direct category.

Forecast reliability and uncertainty

Development-stage forecasts require caution. I treated Blue Ember's contemporaneous plans as evidence of expected economic activity, not as guarantees. Reliability increases because the key forecast predates the excursion, uses an established budgeting cycle, identifies specific milestones, and was relied upon for staffing and vendor commitments.

I tested actual results available through June 2025 only to identify unrelated events and obvious model errors. I did not replace the pre-event perspective with hindsight success. Later performance cannot prove that every expectation was certain on September 12.

The model incorporates technical and program risk through the same probability framework Blue Ember used internally. I excluded an unapproved expansion scenario and amounts beyond the affected horizon. I also subtracted variable costs Blue Ember did not incur during the shift.

Granite Heron may reasonably challenge the risk factor or whether a particular milestone was within contractual contemplation. Those disputes can be evaluated through sensitivity. They do not show that a development-stage schedule necessarily has zero economic value. My 2 million opinion reflects a conservative bounded estimate supported by records rather than an enterprise valuation.

Relationship between event and claimed period

My causation analysis is economic and assumes the quality and fact evidence connects rejection to the excursion. The original validation lot was unavailable for its planned purpose after September 16. Blue Ember then incurred replacement work and used the March 3 Aurora Quay campaign.

I reviewed other program events to avoid attributing independent delay to transportation. No preexisting event in the records required the same replacement interval before September 12. Later scientific and business risks were included in the forecast adjustment rather than charged as certain carrier-caused loss.

The actual schedule does not mean every day until March produced equal damages. My model maps affected milestones and contribution consequences, subtracting periods and costs not economically linked to the replacement timing. It ends at the defined recovery point rather than continuing through trial.

Whether Blue Ember should have used Cobalt Finch is a separate mitigation question. It can reduce otherwise caused delay if the factfinder determines the alternative was reasonable and likely to succeed. It does not change the historical cost of the original lot or whether the Fletcher event occurred.

Assessment of Cobalt Finch offer

The offer identified real capacity and a plausible accelerated path. It was not fictitious, and Cobalt Finch had relevant general capability. At the October 4 deadline, however, the facility had never manufactured BE-214 and program-specific qualification, method planning, document transfer, material readiness, and change control remained incomplete.

The proposed schedule required immediate commitment and overlapping work. Cobalt Finch characterized the project as feasible with elevated schedule risk and did not assign a success probability. A readiness failure after reservation could consume money and scarce Blue Ember personnel while leaving the March option necessary.

Blue Ember engaged during the forty-eight-hour window, asked technical and commercial questions, and documented unresolved items. Its decision favored an established Aurora Quay path. I do not opine that quality rules legally prohibited accepting Cobalt Finch; the evidence shows a genuine judgment under uncertainty.

Because capacity and feasibility are not equivalent to certain successful output, I cannot treat the full 1.8 million claimed avoidance as an undisputed economic saving. A factfinder may weigh probability and choose a partial deduction if supported by the record and instructions.

Mitigation calculation framework

A net mitigation calculation compares loss likely avoided with incremental cost and additional risk caused by the alternative. Granite Heron's model begins with 1.8 million dollars in allegedly avoided delay and subtracts 300,000 dollars for transfer and qualification, yielding 1.5 million dollars.

That subtraction is mathematically correct if both inputs are accepted. The disputed premise is that a timely successful December campaign would avoid the full 1.8 million. The proposal contemplated December 2 manufacturing start, not immediate final validation release, and it contained readiness contingencies.

I constructed a transparent formula: net deduction equals the factfinder's supported avoided-loss amount minus reasonable incremental alternative cost, floored at zero. The calculation should not subtract a separate risk penalty twice if risk has already reduced the avoided-loss input.

Likewise, the 300,000 dollars should not be added to Blue Ember's actual gross damage because it was not incurred. It becomes relevant only when evaluating the hypothetical alternative. If the factfinder concludes no reasonable alternative was established, the gross 4.2 million remains my opinion. If it finds partial feasibility, the workpaper accepts a partial avoided-loss figure.

Response to the proposed 1.5 million deduction

Granite Heron's deduction treats the Cobalt Finch path as though booking, readiness, production, testing, and schedule benefit occurred with sufficient certainty. Dr. Calder's testimony is more qualified: she confirms a real slot and capable organization but also incomplete program approval, potential cancellation expense, elevated schedule risk, and no guaranteed outcome.

Dr. Elodie Park may use expected-value or scenario methods. Such methods require a supported probability, correlation with delay avoided, and consistent treatment of cost. The contemporaneous record does not provide a numerical success probability, so selecting one demands expert judgment and sensitivity rather than a factual certainty.

Granite Heron's 1.8 million avoidance input is derived from Blue Ember's timing model. It should retain the same risk and contribution assumptions used to create the underlying 2 million amount. Mixing an unadjusted optimistic alternative with a risk-adjusted actual loss would be inconsistent.

I therefore reject 1.5 million as a mandatory economic deduction. I do not say the alternative has zero probative value. The offer, response window, capability, unresolved work, estimated cost, and schedule can support a factfinder's reasoned partial assessment.

Sensitivity analysis

If the factfinder accepts no mitigation deduction, total loss is 4.2 million dollars. If it finds 300,000 dollars of net avoidable loss, the resulting total is 3.9 million. If it accepts Granite Heron's complete 1.5 million net deduction, total loss becomes 2.7 million.

Those scenarios are calculated against the same 4.2 million gross base. They do not alter the 2.2 million direct component because the asserted Cobalt Finch benefit concerns timing, not revival of the rejected lot. A different factual finding about direct cost should be entered separately.

For illustration, an avoided-loss finding of 900,000 dollars combined with 300,000 dollars of incremental cost produces a 600,000-dollar net deduction and 3.6 million total. That illustration is not my predicted verdict; it shows how the model avoids double counting and preserves the factfinder's role.

The parties should describe consistently whether a number is gross avoidance, incremental cost, or net deduction. Calling the 300,000-dollar cost an "offset" without context could reverse the arithmetic. My schedules display all three fields and reconcile the final result.

Method limitations

My opinions depend on the completeness and accuracy of Blue Ember's business records, third-party invoices, program evidence, and testimony. I tested material amounts but did not audit the financial statements under assurance standards. An expert damages engagement differs from a financial-statement audit.

The development forecast contains uncertainty even after conservative adjustments. The factfinder may choose a different risk factor or conclude the contract does not permit the time-related category. The schedule exposes those choices rather than hiding them inside one total.

I am not qualified to decide whether the temperature excursion rendered the lot scientifically unsuitable. I relied on the documented disposition and other testimony for that assumption. If the Court or jury rejects causal suitability, the corresponding economic loss must be reconsidered.

I am also not predicting whether Cobalt Finch would have succeeded. The record supports multiple scenarios, and the absence of a contemporaneous numerical probability limits precision. My opinion addresses why Granite Heron's full deduction is not an established economic certainty and how a supported finding can be calculated.

Opinions stated to a reasonable professional degree

First, Blue Ember's source records support 1.35 million dollars of cost embodied in validation lot BE214-VL-2409A. Second, the records support 850,000 dollars of incremental replacement and directly associated expense after reasonable exclusions. Together those direct categories equal 2.2 million dollars.

Third, contemporaneous forecasts and a bounded contribution method support 2 million dollars in delay loss, with technical and program uncertainty reflected and indefinite future sales excluded. Fourth, direct cost and delay are separable components that can be adjusted individually.

Fifth, Cobalt Finch's October offer was real and potentially beneficial but remained subject to program-specific transfer, qualification, readiness, and execution risk. The evidence does not establish Granite Heron's 1.8 million gross avoidance as certain. Sixth, if the factfinder adopts an avoided-loss amount, it should subtract reasonable incremental alternative cost once to obtain a net deduction.

The gross total is 4.2 million dollars. A 300,000-dollar net deduction yields 3.9 million, and a 1.5 million net deduction yields 2.7 million. These calculations are arithmetic consequences, not opinions about which verdict the law requires.

Certification and supporting schedules

I certify that this report states all opinions presently expected within my assigned scope, together with their principal bases. I may supplement if new information materially changes an analysis, respond to another expert, or prepare demonstrative summaries that do not introduce an undisclosed opinion.

The accompanying fictional schedules are: Schedule A, source list; Schedule B, lot-cost reconciliation; Schedule C, replacement-cost reconciliation; Schedule D, pre-event forecast comparison; Schedule E, bounded delay model; Schedule F, Cobalt Finch scenarios; Schedule G, credits review; and Schedule H, qualifications, publications, compensation, and prior testimony.

My electronic workpapers retain formulas and source links in a nonpublic exercise file. Any portable rendering should be understood as a human-readable presentation, not independent machine analysis. I will preserve the version underlying this report.

I declare that the opinions are my own and that I understand my duty to provide accurate, supportable testimony within the permitted expert scope.

Kellan Brooks, CPA/CFF

Fictional forensic accounting expert

Dated July 7, 2025

Every credential, engagement, amount, company, document, and event in this report is fabricated solely for the synthetic appellate record.